Packets

why data travels in small pieces

Watch the 8-min video

In one line: data crosses the network in small packets, each a header (addresses and a number) plus a payload; the MTU, usually 1,500 bytes, sets the size, and the numbers let the receiver reassemble everything and ask again for just the missing piece.

WORDS YOU'LL MEET TODAY

- **Packet**: one small piece of data that travels on its own
- **Payload**: the data inside the packet
- **Reassemble**: put the pieces back together, in order
- **Router**: a machine that passes packets along toward their destination
- **Header**: the label on a packet: where from, where to, its number
- **MTU (maximum transmission unit)**: the largest packet size; 1,500 bytes on most networks
- **Byte**: a tiny unit of data, about one letter of text
- **Megabyte (MB)**: about a million bytes; a phone photo is about 3 MB

1 The question: why pieces?

You're on a train and a friend sends you a photo on WhatsApp. The download circle creeps round, bit by bit: the photo is not arriving all at once, it is arriving in pieces. On the way one piece is lost and two arrive in the wrong order, and yet the photo that opens is perfect. Why cut a photo into pieces at all, and how does your phone put them back together?

One unit first: a **byte** is a tiny unit of data, about one letter of text. A phone photo is around **3 MB**, about three million bytes.



Figure 1. A photo travels as packets: cut into pieces, each labelled with a header, sent on its own, and reassembled by number at the other end.

2 An everyday picture first: moving house

You're moving from Bengaluru to Pune, and your sofa is too big for the courier, whose rule is: nothing bigger than one box. So you take the sofa apart and pack it into 20 boxes, each with a label: from your old flat, to your new flat, box 3 of 20. The boxes travel in different vans on different roads and arrive out of order; box 9 may come before box 2. At the new flat you read the labels, sort the boxes 1 to 20 and put the sofa back together. If box 7 never arrives, you see the gap and ask for box 7 again: only box 7, not the whole sofa.



Boxes go in different vans and arrive out of order (box 9 before box 2).

At the new flat: read the labels, sort 1 to 20, rebuild the sofa (reassemble).

Box 7 missing? You see the gap and ask for box 7 again: only box 7, not the whole sofa.

Sofa = your photo · box = packet · label = header · contents = payload · courier's size rule = MTU

Figure 2. The everyday picture: moving a sofa as 20 numbered boxes.

3 Now the real thing

Moving house	In a network
The sofa	your photo (any data: a message, a web page)
One box	a **packet**: one small piece of the photo
The courier's size rule	the **MTU**: the largest packet size
The label on the box	the **header**: where it comes from, where it's going, its number
What's inside the box	the **payload**: a piece of the photo
Vans and roads	the network
You, sorting boxes at the new flat	the receiver, which **reassembles** the photo

4 Step by step: one photo

1. **Cut.** The sender's phone cuts the photo into pieces, each small enough to fit the size limit. Job: make every piece fit.
2. **Label.** Each piece gets a header in front: from this phone, to the WhatsApp server, and a number. Job: let each piece travel on its own and still find its place.
3. **Send.** The packets leave one after another and each makes its own way across the networks; some take a longer road. The network's job is to carry each packet, nothing more.
4. **Reassemble.** At the other end the packets are sorted by number and joined back into the photo. Job: rebuild the exact original, byte for byte.

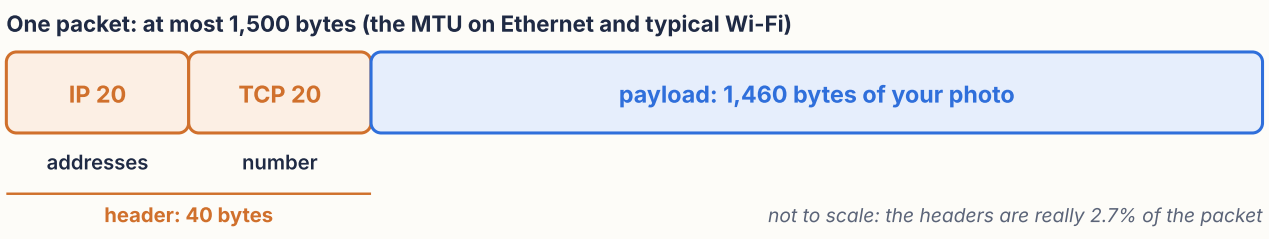


Figure 3. Inside one packet: a 40-byte header (20 bytes IP, 20 bytes TCP) and up to 1,460 bytes of payload.

5 Real numbers

On most networks, including Ethernet and typical Wi-Fi, the largest packet size, the **MTU**, is **1,500 bytes**. Not all of it is photo: 20 bytes are the **IP header** (the addresses, lesson 3) and 20 bytes the **TCP header** (the numbering, lesson 8), which leaves **1,460 bytes** of payload.

Question	Arithmetic	Answer
payload per packet	1,500 − 20 − 20	**1,460 bytes**
packets for a 3 MB photo	3,000,000 ÷ 1,460	**≈ 2,055 packets**
cost of the headers	40 ÷ 1,500	**≈ 2.7%** (under 3%)
resend after one loss	one packet's payload	**1,460 bytes**, not 3,000,000
time for one packet on 10 Mbit/s	1,500 × 8 bits ÷ 10,000,000 bits/s	**1.2 ms**
the photo as one piece on 10 Mbit/s	3,000,000 × 8 ÷ 10,000,000	**2.4 s**

So why pay for the labels? **Loss:** one lost piece costs 1,460 bytes to resend, about two thousand times less than the whole photo. **Sharing:** your family shares one Wi-Fi; while your brother streams a cricket match and you send the photo, the small packets take turns on the wire and nobody waits long.



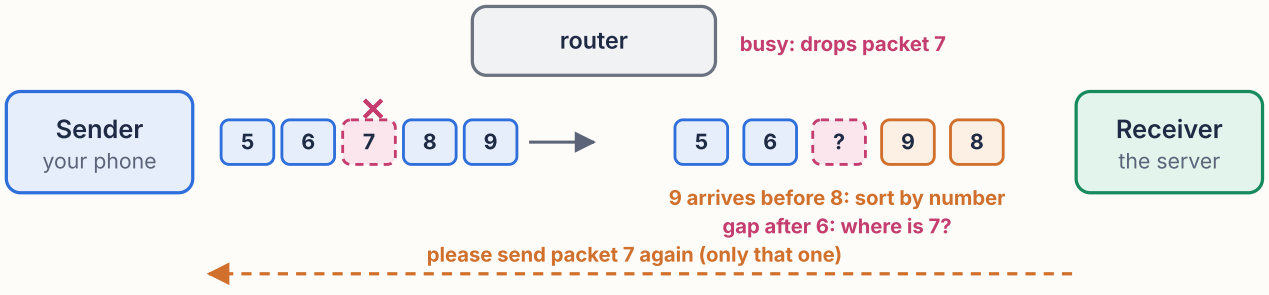
Figure 4. Real numbers: a 3 MB photo is about 2,055 packets; a loss costs one packet; small packets let everyone take turns.

For the curious. 1,500 bytes is the Ethernet limit for the packet itself; the Ethernet frame around it adds a few more bytes that are not counted. Headers can be longer than 20 bytes when options are used, and IPv6 headers are 40 bytes, so real payloads are often a little under 1,460. Strictly, the number in the TCP header counts bytes, not packets; lesson 8 shows how.

6 What goes wrong: lost and out of order

Packet 7 is lost: a busy **router**, the machine that passes packets along, had no room and dropped it. And packets 8 and 9 arrive the wrong way round.

- **Out of order is easy.** The receiver sorts by number, and 8 goes before 9. No harm done.
- **The loss is the real problem.** The receiver sees a gap, 6 then 8: where is 7? It holds on to 8 and 9, and the sender is asked to send packet 7 again. Just that one.
- Packet 7 arrives, the gap is filled, and the photo is rebuilt perfectly. That's why your photo is fine even when the network isn't. Exactly how the asking works is lesson 8.



The receiver holds 8 and 9, gets the new copy of 7, and rebuilds the photo perfectly. How the asking works: lesson 8.

Figure 5. What goes wrong: packet 7 is dropped and 8 and 9 arrive the wrong way round. The receiver sorts by number, sees the gap and asks for 7 again.

PAUSE AND THINK

A friend sends you a voice note of 1.46 MB. Each packet carries 1,460 bytes of payload. How many packets is that? And if packet 500 is lost, what is sent again? Cover the answer and decide.

Answer. 1,460,000 ÷ 1,460 = **1,000 packets**. If packet 500 is lost, only packet 500 is sent again: 1,460 bytes. Your phone keeps packets 501 onwards, waits for 500, and then reassembles the voice note.



Figure 6. The pause-and-think answer: 1,000 packets, and only packet 500 is sent again.

TWO COMMON MISTAKES

- **"A packet is the whole message."** No. A photo or a web page is usually many packets. A packet is one piece, at most 1,500 bytes on most networks.
- **"One lost packet means the whole download starts again."** No. Only the missing piece is sent again; the other pieces wait until the gap is filled.

RECAP IN THREE LINES

1. Data travels in small packets, each a header plus a payload.
2. The MTU, usually 1,500 bytes, makes a 3 MB photo about two thousand packets.
3. Numbers in the headers let the receiver reassemble them, and ask again for just the missing piece.

7 Interview questions: what to say in 30 seconds

Q1. What is a packet, and what is inside one? BASIC

A packet is one small piece of data that travels across the network on its own. It has a header, the label with the source and destination addresses and a number, and a payload, the piece of data itself. A photo or a web page is usually many packets.

Q2. What is the MTU, and what is it on most networks? BASIC

The maximum transmission unit: the largest packet a network link will carry. On Ethernet, and on typical home Wi-Fi, it is 1,500 bytes. After a 20-byte IP header and a 20-byte TCP header, that leaves 1,460 bytes of data per packet.

Q3. How many packets does a 3 MB photo need? SCENARIO

3,000,000 bytes ÷ 1,460 bytes per packet ≈ 2,055 packets, about two thousand. The headers cost 40 bytes in every 1,500, under 3%.

Q4. Why not send the whole file as one big piece? SCENARIO

Two reasons. Loss: if one piece is lost, only 1,460 bytes are resent instead of the whole file. Sharing: small packets take turns on a link, so one big transfer does not block everyone else; on a 10 Mbit/s link a 1,500-byte packet takes 1.2 ms, while a 3 MB file as one piece would hold the link for 2.4 s.

Q5. Packets arrive out of order and one is missing. Is the file broken? TRICKY

No. The receiver sorts the packets by the number in their headers, holds the ones after the gap, and the missing one is sent again. Only when the gap is filled is the data handed over, complete and in order.

Q6. Does one lost packet mean the download restarts? TRICKY

No. Only the missing packet is resent; everything that arrived is kept. (With TCP, lesson 8, the number actually counts bytes, not packets, and the receiver's replies tell the sender what to resend.)

Read more. RFC 791 (Internet Protocol), RFC 9293 (TCP) and RFC 894 (IP over Ethernet, the 1,500-byte limit). The full written lesson is at backendroadmap.com.

Next: SD-N-003 · IP addresses: the street address of every device backendroadmap.com